

16

Conformational analysis

Connections

➡ Building on

- How to determine a molecule's structure [ch3 & ch13](#)
- How some molecules can exist as stereoisomers [ch14](#)

Arriving at

- If I could see a molecule, what would its three-dimensional shape (conformation) be?
- What effect does a molecule's shape have on its reactions?
- How single bonds are free to rotate, but spend most of their time in just two or three well-defined arrangements
- How rings of atoms are usually not planar, but 'puckered'
- How 'puckered' six-membered rings have the most well-defined arrangements of atoms
- How to draw six-membered rings accurately
- How to use the known arrangements of the atoms in a six-membered ring to predict and explain their reactions

➡ Looking forward to

- How conformation, and the alignment of atoms, can affect elimination reactions [ch17](#)
- How NMR spectroscopy backs up what we have said in this chapter [ch31](#)
- How the conformation of molecules dictates how they react, e.g. from which direction they will be attacked by reagents [ch32 & ch33](#)
- How the alignment of bonds can allow groups in molecules to move around (rearrangement reactions) or allow C–C bonds to break (fragmentation reactions) [ch36](#)
- How the alignment of orbitals controls reactivity (stereoelectronics) [ch31](#)
- The accurate drawing of rings as transition states is necessary [ch32, ch34, & ch35](#)

Bond rotation allows chains of atoms to adopt a number of conformations

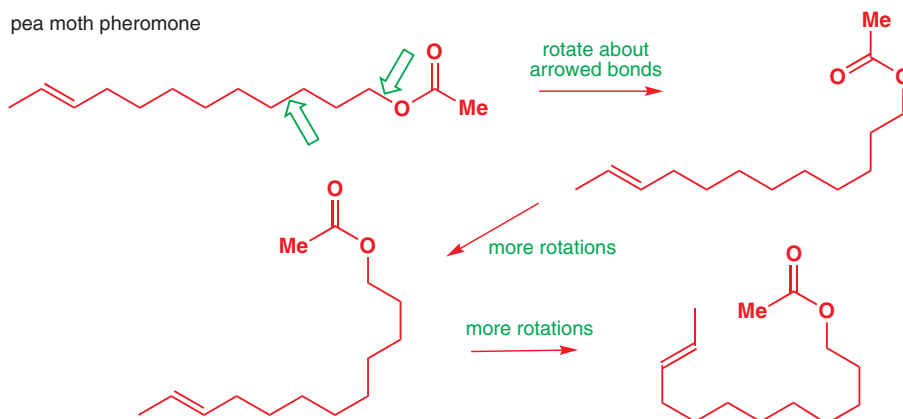
Several chapters of this book have considered how to find out the structure of molecules. We have seen X-ray crystallography pictures, which reveal exactly where the atoms are in crystals; we have looked at IR spectroscopy, which gives us information about the bonds in the molecule, and at NMR spectroscopy, which gives us information about the atoms themselves and how they are joined up. Up to now, we have mainly been interested in determining which atoms are bonded to which other atoms and also the shapes of small localized groups of atoms. For example, a methyl group has three hydrogen atoms bonded to one carbon atom and the atoms around this carbon are located at the corners of a tetrahedron; a ketone consists of a carbon atom bonded to two other carbon atoms and doubly bonded to an oxygen atom, with all these atoms in the same plane.

But, on a slightly larger scale, shape is not usually so well defined. Rotation is possible about single bonds and this rotation means that, while the localized arrangement of atoms stays the same (every saturated carbon atom is still always tetrahedral), the molecule as a whole can adopt a number of different shapes. Shown on the next page are several snapshot views of one molecule—it happens to be a pheromone used by pea moths to attract a mate. Although the structures look dissimilar, they differ from one another only by rotation about one or more single bonds. Whilst the overall shapes differ, the localized structure is still the

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1, 123-2** etc. (replacing **123** with the page number) for access to successive links.

same: tetrahedral sp^3 carbons; trigonal planar sp^2 carbons. Notice another point too, which we will pick up on later: the arrangement about the double bond always remains the same because double bonds can't rotate.

■ This point is also discussed in Chapters 4 and 12.



Interactive conformations of pea moth pheromone

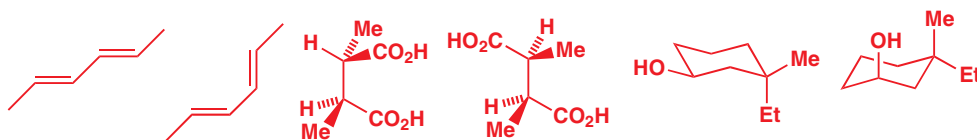
At room temperature in solution, all the single bonds in the molecule are constantly rotating—the chances that two molecules will have exactly the same shape at any one time are quite small.

Yet, even though no two molecules have exactly the same shape at any one time, they are still all the same chemical compound—they have all the same atoms attached in the same way. We call the different shapes of molecules of the same compound different **conformations**.

Conformation and configuration

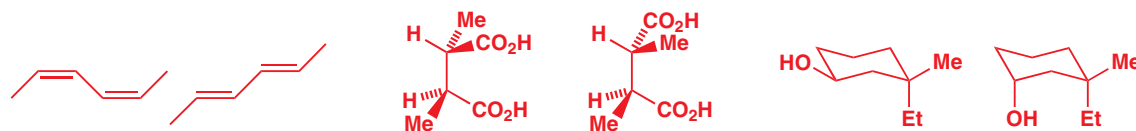
To get from one conformation to another, we can rotate about as many single bonds as we like. The one thing we can't do though is to break any bonds. This is why we can't rotate about a double bond—to do so we would need to break the π bond. Below are some pairs of structures that can be interconverted by rotating about single bonds: they are all different conformations of the same molecule.

three compounds, each shown in two conformations



The next block of molecules is something quite different: these pairs can only be interconverted by breaking a bond. This means that they have different **configurations**—configurations can be interconverted only by breaking bonds. Compounds with different configurations are called **stereoisomers** and we dealt with them in Chapter 14.

three pairs of stereoisomers: each member of a pair has a different configuration



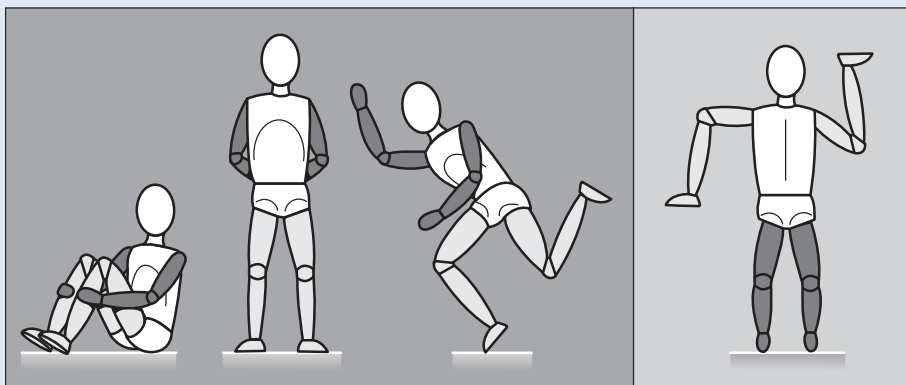
● Rotation or bond breaking?

- Structures that can be interconverted simply by rotation about single bonds are **conformations** of the same molecule.
- Structures that can be interconverted only by breaking one or more bonds have different **configurations**, and are stereoisomers.

Conformation and configuration

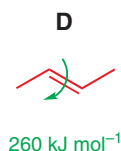
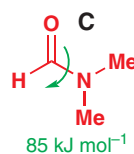
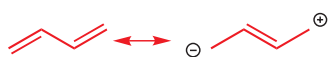
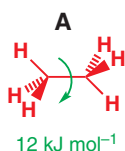
Different conformations of a person – Some more stable than others ...

A different configuration



Barriers to rotation

We saw in Chapter 7 that rotation about the C–N bond in an amide is relatively slow at room temperature—the NMR spectrum of DMF clearly shows two methyl signals (p. 156). In Chapter 12 you learned that the rate of a chemical process is associated with an energy barrier (this holds both for reactions and simple bond rotations): the lower the rate, the higher the barrier. Rotation about single bonds is fast at room temperature, but there is nonetheless a barrier to rotation in ethane (A), for example, of about 12 kJ mol^{-1} .



The energy barrier for rotation about the single bond in butadiene (B) is slightly larger because of the weak conjugation between the double bonds, but the barrier to rotation about the genuine double bond in but-2-ene (D) is enormous and no rotation is seen. The energy barrier to the rotation about the C–N bond in an amide such as DMF (C) is usually about 80 kJ mol^{-1} , translating into a rate of about 0.1 s^{-1} at 20°C . The conjugation in amides is well developed, and the C–N bond has significant double-bond character.

Rates and barriers

It can be useful to remember some simple guidelines to the way in which energy barriers relate to rates of rotation, as discussed in Chapter 12. For example:

- A barrier of 73 kJ mol^{-1} allows one rotation every second at 25°C (that is, the rate is 1 s^{-1}).
- Every 6 kJ mol^{-1} changes the rate at 25°C by about a factor of 10.

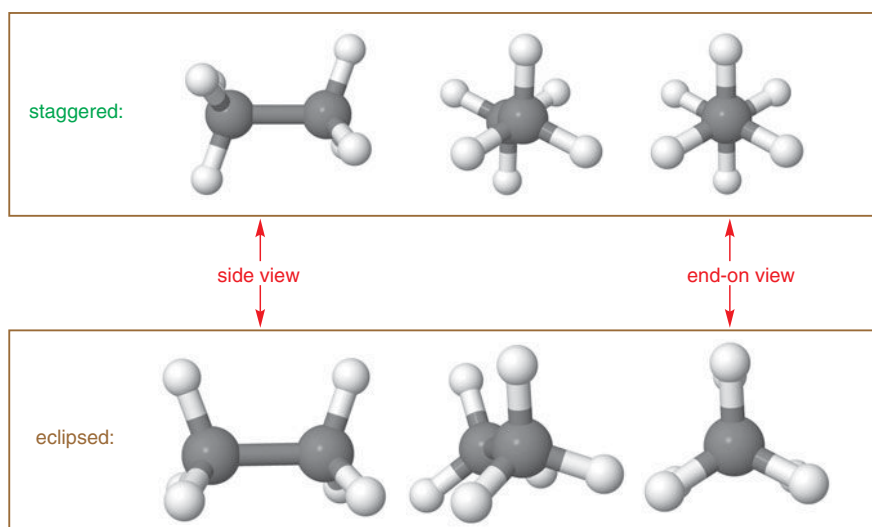
To see signals in an NMR spectrum for two different conformations, they must interconvert no faster than (very roughly) 1000 s^{-1} —a barrier of about 55 kJ mol^{-1} at 25°C . This is why NMR shows two methyl signals for DMF, but only one set of signals for butadiene. See p. 374 for more on this.

For conformations to interconvert slowly enough for them to exist as different compounds, the barrier must be over 100 kJ mol^{-1} . The barrier to rotation about a $\text{C}=\text{C}$ double bond is 260 kJ mol^{-1} —which is why we can separate *E* and *Z* isomers.

Conformations of ethane

Why should there be an energy barrier in the rotation about a single bond? In order to answer this question, we should start with the simplest $\text{C}-\text{C}$ bond possible—the one in ethane. Ethane has two extreme conformations called the staggered and eclipsed conformations. Three different views of these are shown below.

the two extreme conformations of ethane, staggered and eclipsed, each shown from three different viewpoints

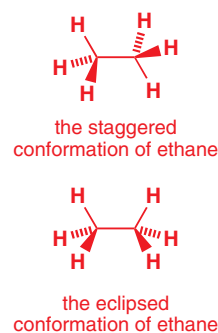


You can see why the conformations have these names by looking at the end-on views in the diagram. In the eclipsed case the near $\text{C}-\text{H}$ bonds completely block the view of the far bonds, just as in a solar eclipse the Moon blocks the Sun as seen from the Earth. In the staggered conformation, the far $\text{C}-\text{H}$ bonds appear in the gaps between the near $\text{C}-\text{H}$ bonds—the bonds are staggered.

Chemists often want to draw these two conformations quickly and two different methods are commonly used, each with its own merits. In the first method, shown on the right, we simply draw the side view of the molecule and use wedged and hashed lines to show bonds not in the plane of the paper (as you saw in Chapter 14). Particular attention must be paid to which of the bonds are in the plane and which go into and out of the plane.

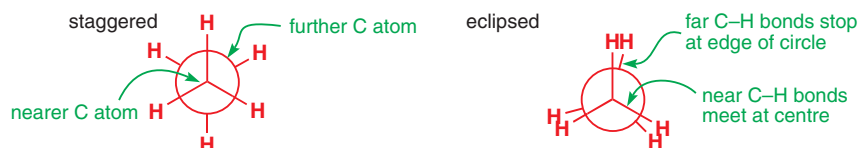
In the second method we draw the end-on view, looking along the $\text{C}-\text{C}$ bond. This view is known as a Newman projection, and Newman projections are subject to a few conventions:

- The carbon atom nearer the viewer is at the junction of the front three bonds.
- The carbon further away (which can't in fact be seen in the end-on view) is represented by a large circle. This makes the perspective inaccurate, but this doesn't matter.



- Bonds attached to this further carbon join the *edge* of the circle and do not meet in the centre.
- Eclipsed bonds are drawn slightly displaced for clarity—as though the bond were rotated by a tiny fraction.

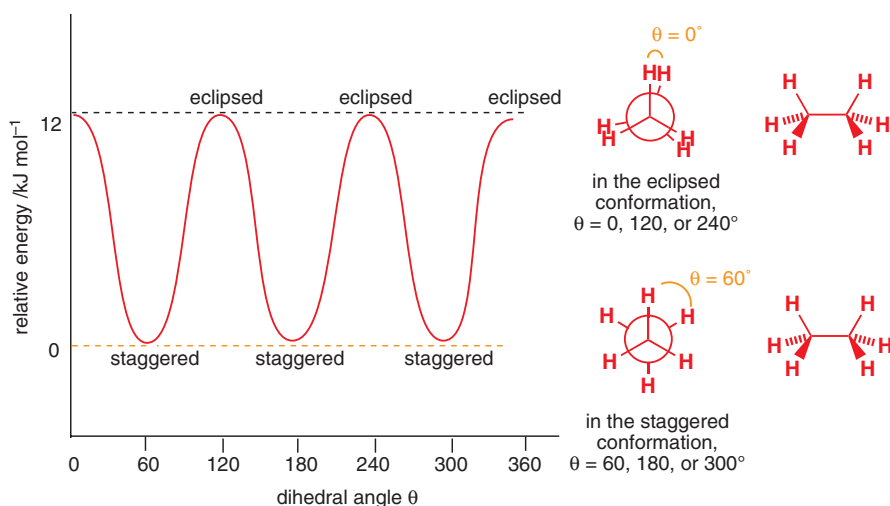
Newman projections of the staggered and eclipsed conformations of ethane



■ Picturing dihedral angles is sometimes hard—one way to do it is to imagine the two C–H bonds drawn on two facing pages of a book. The dihedral angle is then the angle between the pages, measured perpendicular to the spine. See Chapter 31.

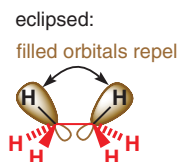
The staggered and eclipsed conformations of ethane are not identical in energy: the staggered conformation is lower in energy than the eclipsed by 12 kJ mol^{-1} , the value of the rotational barrier. Of course, there are other possible conformations too with energies in between these extremes, and we can plot a graph to show the change in energy of the system as the C–C bond rotates. We define the dihedral angle, θ (sometimes called the torsion angle), to be the angle between a C–H bond at the nearer carbon and a C–H bond at the far carbon. In the staggered conformation, $\theta = 60^\circ$ whilst in the eclipsed conformation, $\theta = 0^\circ$.

The energy level diagram shows the staggered conformation as a potential energy minimum whilst the eclipsed conformation represents an energy maximum. This means that the eclipsed conformation is not a stable conformation since any slight rotation will lead to a conformation lower in energy. The molecule will actually spend the vast majority of its time in a staggered or nearly staggered conformation and only briefly pass through the eclipsed conformation *en route* to another staggered conformation.



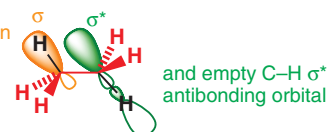
Interactive conformations of ethane

But *why* is the eclipsed conformation higher in energy than the staggered conformation? There are two reasons. The first is that the electrons in the bonds repel each other and this repulsion is at a maximum when the bonds are aligned in the eclipsed conformation. The second is that there may be some stabilizing interaction between the C–H σ bonding orbital on one carbon and the C–H σ^* antibonding orbital on the other carbon, which is greatest when the two orbitals are exactly parallel: this only happens in the staggered conformation. The same effects—repulsion between filled orbitals (a form of steric effect, see p. 129) and stabilization by donation into antibonding orbitals—govern the favoured conformations about all rotating bonds.



staggered:

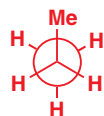
stabilizing interaction between filled C–H σ bond...



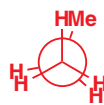
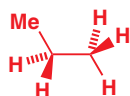
Conformations of propane

Propane is the next simplest hydrocarbon. Before we consider what conformations are possible for propane we should first look at its geometry. The C—C—C bond angle is not 109.5° (the tetrahedral angle, see Chapters 2 and 4) as we might expect, but 112.4° . Consequently, the H—C—H bond angle on the central carbon is smaller than the ideal angle of 109.5° , only 106.1° . This does not necessarily mean that the two methyl groups on the central carbon clash in some way, but instead that two C—C bonds repel each other more than two C—H bonds do.

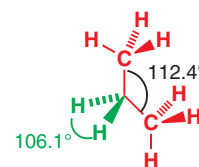
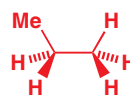
As in the case of ethane, two extreme conformations of propane are possible—in one the C—H and C—C bonds are staggered, in the other they are eclipsed.



the staggered conformation of propane



the eclipsed conformation of propane



there is greater repulsion between two C—C bonds than between two C—H bonds

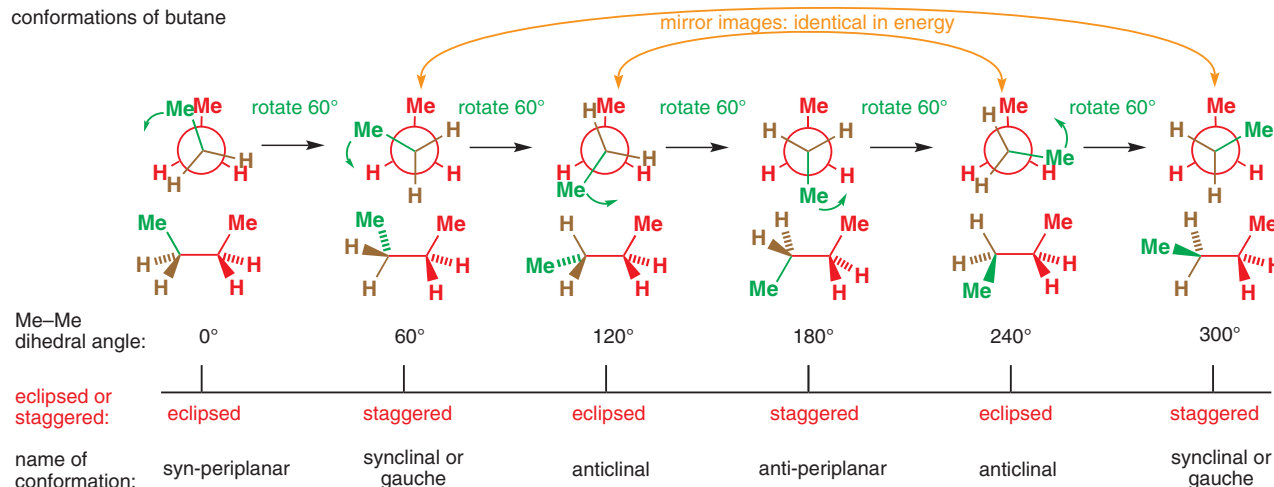
■ Notice that when we draw the eclipsed conformation we have to offset the front and back bonds slightly to see the substituents clearly. In reality, one is right behind the other.

The rotational barrier is now slightly higher than for ethane: 14 kJ mol^{-1} as compared to 12 kJ mol^{-1} . This again reflects the greater repulsion of electrons in the coplanar bonds in the eclipsed conformation rather than any steric interactions. The energy graph for bond rotation in propane would look exactly the same as that for ethane except that the barrier is now 14 kJ mol^{-1} .

Conformations of butane

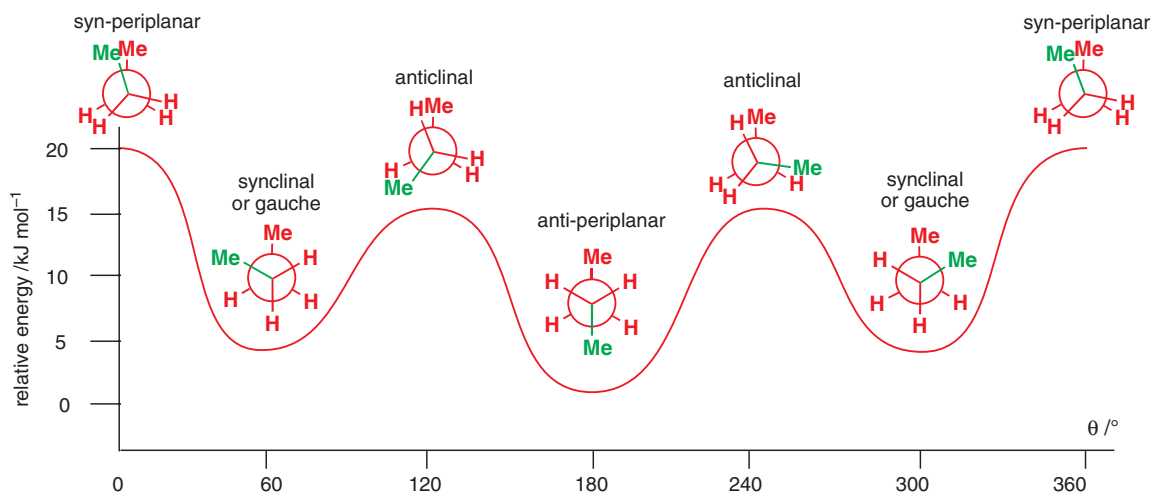
With butane things start to get slightly more complicated. Now we have effectively replaced two hydrogen atoms in ethane by larger methyl groups. These are large enough to get in the way of each other, and steric repulsion becomes a significant contribution to the rotational energy barriers. However, the main complication is that, as we rotate about the central C—C bond, not all the staggered conformations are the same, and neither are all the eclipsed conformations. The six conformations that butane can adopt as the central C—C bond is rotated in 60° intervals are shown below. The green Me group and the brown hydrogens are rotating while the substituents on the other carbon atom remain still.

conformations of butane



Look closely at these different conformations. The conformations with dihedral angles 60° and 300° are actually mirror images of each other, as are the conformations with angles 120° and 240° . This means that we really only have four different maxima or minima in energy as we rotate about the central C—C bond: two types of eclipsed conformation, which will represent maxima in the energy-rotation graph, and two types of staggered conformation, which will represent minima. These four different conformations have names, shown in the bottom row of the diagram. In the syn-periplanar and anti-periplanar conformations the two C—Me bonds lie in the same plane; in the synclinal (or gauche) and anticlinal conformations they slope towards (*syn*) or away from (*anti*) one another.

Before we draw the energy–rotation graph, let’s just stop and think what it might look like. Each of the eclipsed conformations will be energy maxima but the syn-periplanar conformation ($\theta = 0^\circ$) will be higher in energy than the two anticlinal conformations ($\theta = 120^\circ$ and 240°): in the syn-periplanar conformation two methyl groups are eclipsing each other whereas in the anticlinal conformations each methyl group is eclipsing only a hydrogen atom. The staggered conformations will be energy minima but the two methyl groups are furthest from each other in the anti-periplanar conformation so this will be a slightly lower minimum than the two synclinal (gauche) conformations.



Interactive conformations of butane

■ The rotation is very rapid indeed: the barrier of 20 kJ mol^{-1} corresponds to a rate at room temperature of $2 \times 10^9 \text{ s}^{-1}$. This is far too fast for the different conformers to be detected by NMR (see p. 363): the NMR spectrum of butane shows only one set of signals representing an average of all the conformations.

■ You will see why such detailed conformational analysis of acyclic compounds is so important in Chapter 17 on eliminations, where the products of elimination reactions can be explained only by considering the conformations of the reactants and the transition states. But first we want to use these ideas to explain another branch of organic chemistry—the conformation of ring structures.

As in ethane, the eclipsed conformations are not stable since any rotation leads to a more stable conformation. The staggered conformations are stable since they each lie in a potential energy well. The anti-periplanar conformation, with the two methyl groups opposite each other, is the most stable of all. We can therefore think of a butane molecule as rapidly interconverting between synclinal and anti-periplanar conformations, passing quickly through the eclipsed conformations on the way. The eclipsed conformations are energy maxima, and therefore represent the transition states for interconversion between conformations.

If we managed to slow down the rapid interconversions in butane (by cooling to very low temperature, for example), we would be able to isolate the three stable conformations—the anti-periplanar and the two synclinal conformations. These different stable conformations of butane are some sort of isomers. They are called *conformational isomers* or *conformers* for short.

● Conformations and conformers

Butane can exist in an infinite number of *conformations* (we have chosen to show only the six most significant) but has only three *conformers* (potential energy minima)—the two synclinal (gauche) conformations and the anti-periplanar conformation.

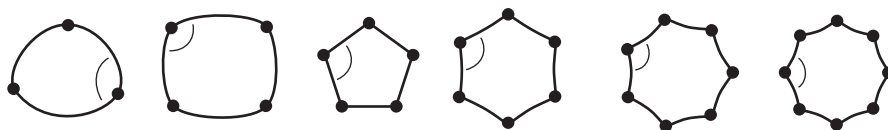
You now have a more thorough explanation of the zig-zag arrangement of carbon chains, first introduced in Chapter 2 when we showed you how to draw molecules realistically. This is the shape you get if you allow all the C–C bonds to take up the anti-periplanar conformation, and it will be the most stable conformation for any linear alkane.

Ring strain

Up to now, we haven’t given an entirely accurate impression of rings. We have been drawing them all as though they were planar, although this is actually not the case. In this section you will learn how to draw rings more accurately and to understand the properties of the different conformations adopted.

If we assume that in fully saturated carbocyclic rings each carbon is sp^3 hybridized, then each bond angle would ideally be 109.5° . However, in a planar ring, the carbon atoms don’t

have the luxury of choosing their bond angles: internal angle depends only on the number of atoms in the ring. If this angle differs from the ideal 109.5° , there will be some sort of strain in the molecule. This is best seen in the picture below, where the atoms are forced planar. The more strained the molecules are, the more the bonds curve—in a strain-free molecule, the bonds are straight.



All internal angles 109.5°

Notice how in the smaller rings the bonds curve outwards, whilst in the larger rings the bonds curve inwards. The table gives values for the internal angles for regular planar polygons and an indication of the strain per carbon atom due to the deviation of this angle from the ideal tetrahedral angle of 109.5° .

These data are best presented as a graph, and the ring strain per carbon atom in planar rings for ring sizes up to 17 are shown on the next page. Whether the bonds are strained inwards or outwards is not important so only the magnitude of the strain is shown.

From these figures (represented in the graph on p. 368), note:

- These are calculated data for hypothetical planar rings. As you will see, real rings are rather different.
- The calculated ring strain is largest for three-membered rings but rapidly decreases through a four-membered ring and reaches a minimum for a five-membered ring.
- The calculated ring strain increases again (although less rapidly) as the rings get larger after the minimum at 5.

But what we really need is a measure of the strain in actual compounds, not just a theoretical prediction in planar rings, so that we can compare this with the theoretical angle strain. A good measure of the strain in real rings is obtained using heats of combustion. Look at the following heats of combustion for some straight-chain alkanes. What is striking is that the difference between any two in the series is very nearly constant at around -660 kJ mol^{-1} .

Heats of combustion for some straight-chain alkanes

Straight-chain alkane	$\text{CH}_3(\text{CH}_2)_n\text{CH}_3$, $n =$	$-\Delta H_{\text{combustion}}$, kJ mol^{-1}	Difference, kJ mol^{-1}
ethane	0	1560	
propane	1	2220	660
butane	2	2877	657
pentane	3	3536	659
hexane	4	4194	658
heptane	5	4853	659
octane	6	5511	658
nonane	7	6171	660
decane	8	6829	658
undecane	9	7487	658
dodecane	10	8148	661

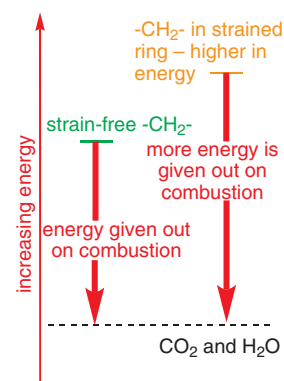
If we assume (as is reasonable) that there is no strain in the straight-chain alkanes, then each extra methylene group, $-\text{CH}_2-$, contributes on average an extra $658.7 \text{ kJ mol}^{-1}$ to the heat of combustion for the alkane. A cycloalkane $(\text{CH}_2)_n$ is simply a number of methylene groups joined together. If the cycloalkane is strain-free, then its heat of combustion should be $n \times 658.7 \text{ kJ mol}^{-1}$. If, however, there is some strain in the ring that makes the ring less stable (that is, raises its energy) then more energy is given out on combustion. Now, let's put all this together in a graph showing, for each ring size: (a) angle strain per CH_2 group and (b) heat of combustion per CH_2 group.

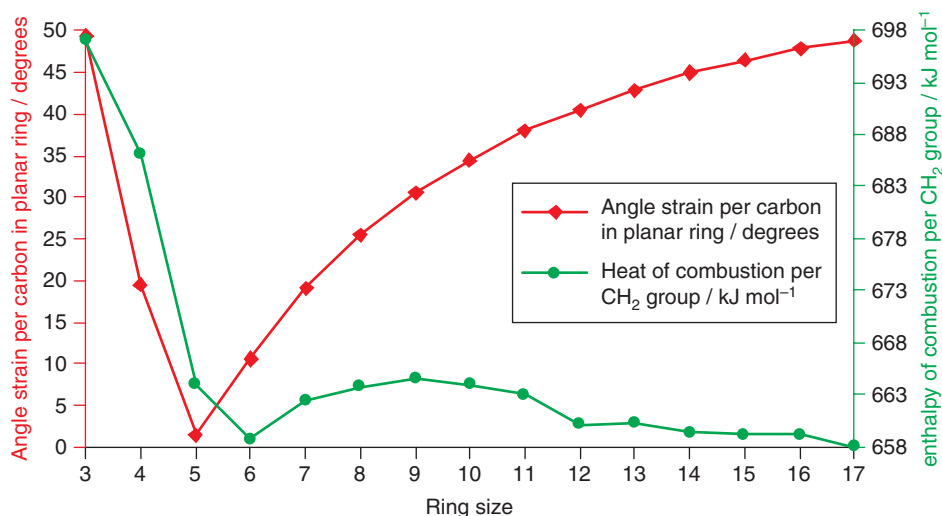
■ We have used ring strain a number of times to explain the reactivity of cyclic molecules (p. 352).

Number of atoms in ring	Internal angle in planar ring	$109.5^\circ - \text{internal angle}^a$
3	60°	49.5°
4	90°	19.5°
5	108°	1.5°
6	120°	-10.5°
7	128.5°	-19°
8	135°	-25.5°

^a A measure of strain per carbon atom.

■ A similar measurement was used in Chapter 7 to demonstrate the stabilization of benzene due to its aromaticity.





Points to notice in graph above:

- The greatest strain by far is in the three-membered ring, cyclopropane ($n = 3$).
- The strain decreases rapidly with ring size but reaches a minimum for cyclohexane, *not* cyclopentane as you might have predicted from the angle calculations.
- The strain then increases but not nearly as quickly as the angle calculation suggested: it reaches a maximum at around $n = 9$ and then decreases once more.
- The strain does not go on increasing as ring size increases but instead remains roughly constant after about $n = 14$.
- Cyclohexane ($n = 6$) and the larger cycloalkanes ($n \geq 14$) all have heats of combustion per $-\text{CH}_2$ group of around 658 kJ mol^{-1} , the same value as that of a $-\text{CH}_2$ group in a straight-chain alkane, that is, *they are essentially strain-free*.

You might ask yourself some questions now:

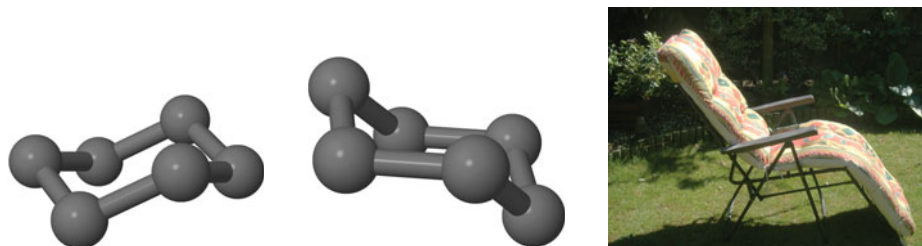
Why are six-membered rings and large rings virtually strain-free?

Why is there still some strain in five-membered rings even though the bond angles in a planar structure are almost 109.5° ?

The answer to both these questions, as you may already have guessed, is that the assumption that the rings are planar is simply not correct. It is easy to see how large rings can fold up into many different conformations as easily as acyclic compounds do. It is less clear to predict what happens in six-membered rings.

Six-membered rings

If you were to join six tetrahedral carbon atoms together, you would probably find that you ended up with a shape like this.



All the carbon atoms are certainly not in the same plane, and there is no strain because all the bond angles are 109.5° . If you squash the model against the desk, forcing the atoms to lie in the same plane, it springs back into this shape as soon as you let go. If you view the model from one side (the second picture above) you will notice that four carbon atoms lie in the same plane, with the fifth above the plane and the sixth below it (although it's important to realize that all six are identical—you can check this by rotating your model). The slightly overly imaginative name for this conformation—the **chair conformation**—derives from this view.

Chemists class rings as small, normal, medium, or large depending on their size.

small, $n = 3$ or 4

normal, $n = 5, 6$, or 7

medium, $n = 8$ to about 14

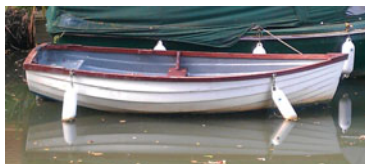
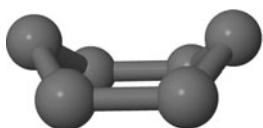
large, $n >$ about 14

This is because these different classes all have different properties and synthetic routes to making them. The groupings are evident in the graph above.

■ By far the easiest way to get to grips with these different shapes is by building models. We strongly recommend you do this!



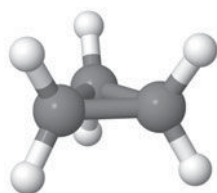
There is another conformation of cyclohexane that you might have made that looks like this.



This conformation is known as the **boat conformation**. In this conformation there are still four carbon atoms in one plane, but the other two are both above this plane. Now all the carbon atoms are not the same—the four in the plane are different from the ones above. However, this is not a stable conformation of cyclohexane, even though there is no bond angle strain (all the angles are 109.5°). In order to understand why not, we must go back a few steps and answer our other question: why is cyclopentane strained even though a planar conformation has virtually no angle strain?

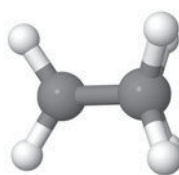
Smaller rings (three, four, and five members)

The three carbon atoms in cyclopropane must lie in a plane since it is always possible to draw a plane through any three points. All the C–C bond lengths are the same, which means that the three carbon atoms are at the corners of an equilateral triangle. From the large heat of combustion per methylene group (p. 368) we know that there is considerable strain in this molecule. Most of this is due to the bond angles deviating so greatly from the ideal tetrahedral value of 109.5° . Most—but not all. If we view along one of the C–C bonds we can see a further cause of strain—all the C–H bonds are eclipsed.

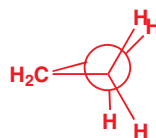


a side-on view of cyclopropane

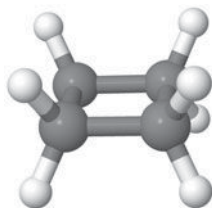
view along C–C



viewing cyclopropane along a C–C bond shows that all the C–H bonds are eclipsed

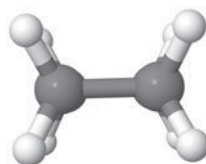


The eclipsed conformation of ethane is an energy maximum and any rotation leads to a more stable conformation. In cyclopropane it is not possible to rotate any of the C–C bonds and so all the C–H bonds are forced to eclipse their neighbours. In fact, in any planar conformation all the C–H bonds will be eclipsed with their neighbours. In cyclobutane, the ring distorts from a planar conformation in order to reduce the eclipsing interactions, even though this reduces the bond angles further and so increases the bond angle strain. Cyclobutane adopts a puckered or 'wing-shaped' conformation.

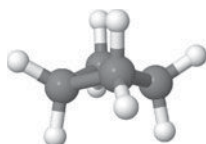


planar cyclobutane (not the real conformation)

view along C–C

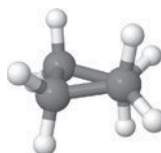


side-on view of planar cyclobutane shows eclipsing C–H bonds

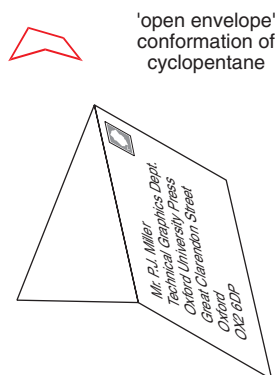


the puckered 'wing' conformation of cyclobutane

view along C–C



C–H bonds no longer fully eclipsed

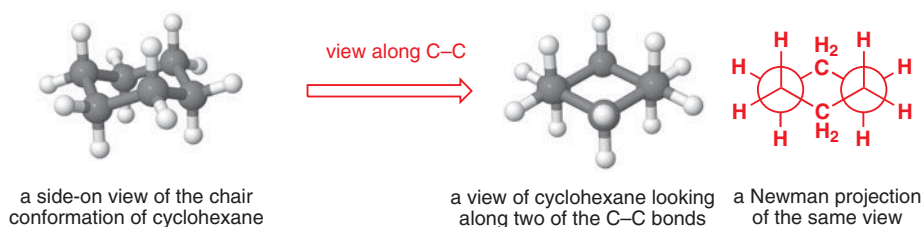


This explains why cyclopentane is not entirely strain-free even though in a planar conformation the C—C—C bond angles are close to 109.5° . The heat of combustion data give us an indication of the total strain in the molecule, not just the contribution of angle strain. There is strain in planar cyclopentane caused by the eclipsing of adjacent C—H bonds. As in cyclobutane, the ring distorts to reduce the eclipsing interactions but this increases the angle strain. Whatever happens, there is always going to be some strain in the system. The minimum energy conformation adopted is a balance of the two opposing effects. Cyclopentane adopts a shape approximating to an 'open envelope', with four C atoms in a plane and one above or below it. The atoms in the ring rapidly take turns not to be in the plane, and cyclopentanes have much less well-defined conformational properties than cyclohexanes, to which we shall now return.

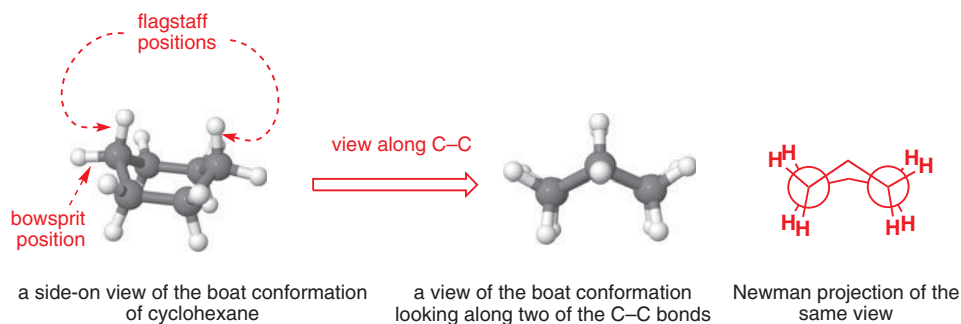
A closer look at cyclohexane

■ We shall consider the conformations, and reactions, of cyclopentanes in Chapter 32.

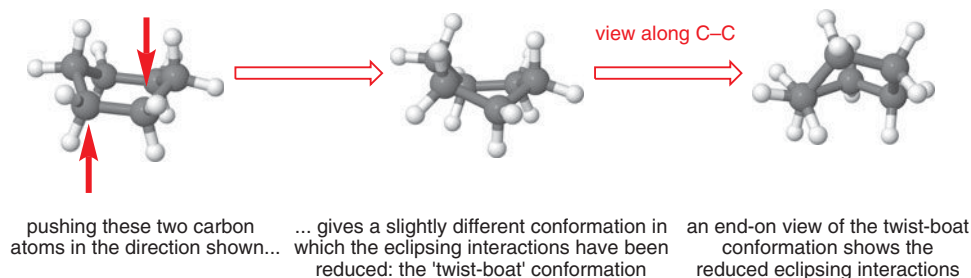
The heats of combustion data on p. 368 show that cyclohexane is virtually strain-free. This must mean that not only is there no angle strain, but there is also no strain from eclipsing interactions either. A model of the chair conformation of cyclohexane including all the hydrogen atoms looks like this.



The view along two of the C—C bonds clearly shows that there are no eclipsing C—H bonds in the chair conformation of cyclohexane—in fact, all the bonds are fully staggered, giving the lowest energy possible. This is why cyclohexane is strain-free. Contrast this with the boat conformation. Now four pairs of C—H bonds are eclipsed, and there is a particularly bad interaction between the 'flagstaff' C—H bonds.



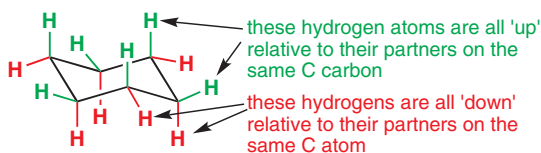
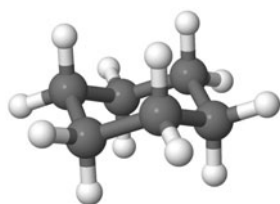
This explains why the boat conformation is much less important than the chair conformation. Even though both are free from angle strain, the eclipsing interactions in the boat conformation make it approximately 25 kJ mol^{-1} higher in energy than the chair conformation. In fact, as we shall see later, the boat conformation represents an energy maximum in cyclohexane whilst the chair conformation is an energy minimum. Earlier we saw how the eclipsing interactions in planar cyclobutane and cyclopentane could be reduced by distortion of the ring. The same is true for the boat conformation of cyclohexane. The eclipsing interactions can be relieved slightly if the two 'side' C—C bonds twist relative to each other.



This twisting gives rise to a slightly different conformation of cyclohexane called the **twist-boat conformation**, which, although not as low in energy as the chair form, is lower in energy (by 4 kJ mol^{-1}) than the boat form and is a local energy minimum, as we shall see later. Cyclohexane has two stable conformers, the chair and the twist-boat. The chair form is approximately 21 kJ mol^{-1} lower in energy than the twist-boat form.

Axial and equatorial

Take another look at the chair conformation on p. 368. All six carbon atoms are identical, but there are two types of protons—one type stick either vertically up or down and are called **axial** hydrogen atoms; the other sort stick out sideways and are called **equatorial** hydrogen atoms. As you go round the ring, notice that each of the CH_2 groups has one hydrogen sticking up and one sticking down. However, all the 'up' ones alternate between axial and equatorial, as do all the 'down' ones.



Before going any further, it's important that you learn how to draw cyclohexane properly. Without cluttering the structure with Cs and Hs, a chemist would draw cyclohexane as one of these three structures.



Up to now, we have simply used the hexagon A to represent cyclohexane. We shall see that, whilst this is the least informative of the three, it is nonetheless still useful. The more informative structures B and C (which are actually just different views of the same molecule) take some practice to draw properly, but you need to be able to draw convincing cyclohexanes and it is worth taking the time to learn how to now.

Guidelines for drawing cyclohexane

The carbon skeleton

Trying to draw the chair conformation of cyclohexane in one continuous line can lead to some dreadful diagrams. The easiest way to draw a chair conformation is by starting off with one end.

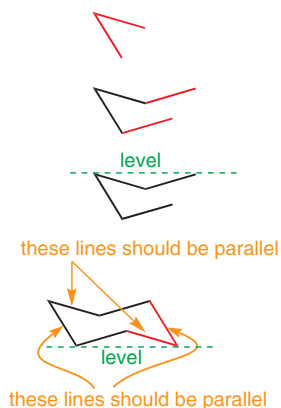
Next draw in two parallel lines of equal length.

At this stage, the top of the new line should be level with the top of the original pair.

Finally, the last two lines should be added. These lines should be parallel to the first pair of lines as shown and the lowest points should also be level.

■ A **local energy minimum** is the bottom of the potential energy well, but not necessarily the deepest possible well, which is the **global energy minimum**. Small changes in conformation will increase the energy, although a large change may be able to decrease the energy further. As an example, the synclinal (gauche) conformation of butane is a *local* energy minimum; the anti-periplanar conformation is the *global* energy minimum.

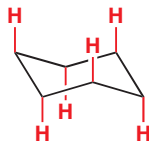
■ Compare the equator and axis of the Earth: equatorial bonds are around the equator of the molecule. Note the spelling (**not** equitorial!).



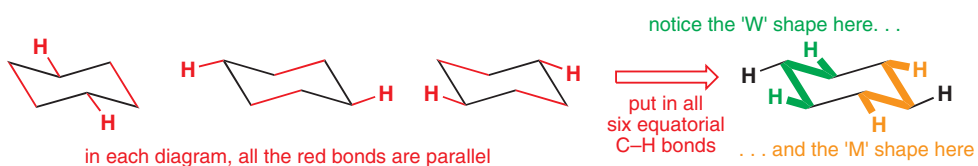
Adding the hydrogen atoms

This is often the trickiest part. Just remember that you are trying to make each of the carbon atoms look tetrahedral. (Note that we don't normally use wedged and hashed bonds in these chair-shaped diagrams; otherwise things get really messy.)

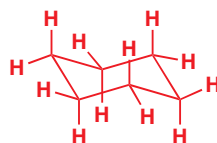
The axial bonds are relatively easy to draw in. They should all be vertically aligned and alternate up and down all round the ring.



The equatorial bonds require a little more care to draw. The thing to remember is that each equatorial bond must be parallel to two C–C bonds.



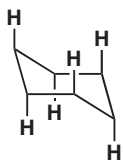
The complete diagram with all the hydrogen atoms should look like this. Most of the time you won't want to draw in *all* the Hs but you need to know how to orientate them in case you do need to.



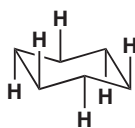
Common mistakes

If you follow all the guidelines above, you will soon be drawing good conformational diagrams. However, a few common mistakes have been included to show you what not to do!

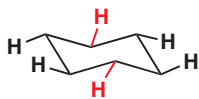
how *not* to draw cyclohexanes...



The chair has been drawn with the middle bonds horizontal, so the upper points of the chair are not level. This means the axial hydrogens can no longer be drawn correctly vertical.



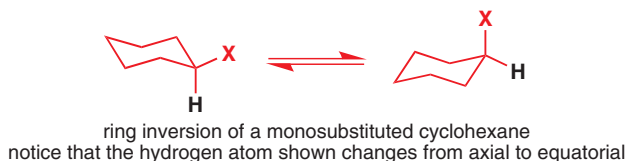
The axial hydrogens have been drawn alternating up and down on the wrong carbons. This structure is impossible because none of the carbons can be tetrahedral.



The red hydrogens have been drawn at the wrong angles—look for the parallel lines and the 'W' and 'M'.

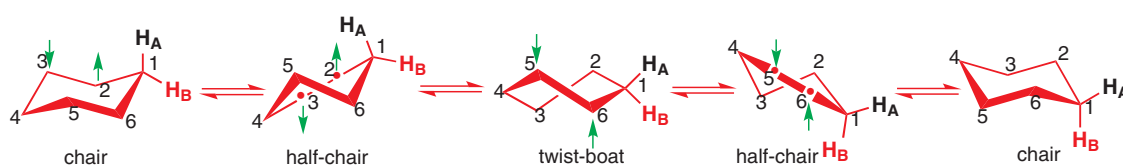
The ring inversion (flipping) of cyclohexane

Given that this chair conformer is the preferred conformation for cyclohexane, what would you expect its ^{13}C NMR spectrum to look like? All six carbon atoms are the same so there should only be one signal (and indeed there is, at 25.2 ppm). But what about the ^1H NMR spectrum? The two different sorts of protons (axial and equatorial) ought to resonate at different frequencies, so two signals should be seen (each with coupling to neighbouring protons). In fact, there is only *one* resonance in the proton spectrum, at 1.40 ppm. In a monosubstituted cyclohexane there should be two isomers detectable—one with the substituent axial, the other with the substituent equatorial. But again at room temperature only one set of signals is seen.

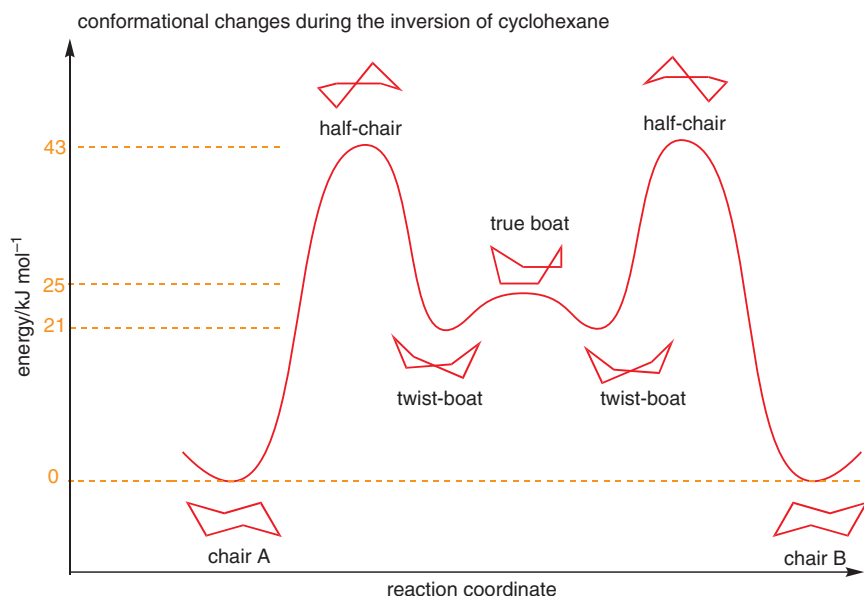


This changes when the NMR spectrum is run at low temperature. Now two isomers are visible, and this gives us a clue as to what is happening: the two isomers are conformers that interconvert—rapidly at room temperature, but more slowly when the temperature is lowered. Recall that NMR does not distinguish between the three different stable conformers of butane (two synclinal and one anti-periplanar) because they are all rapidly interconverting so fast that only an average is seen. The same happens with cyclohexane—just by rotating bonds (that is, without breaking any!) cyclohexane can ring invert or ‘flip’. After ring inversion has taken place, all the bonds that were axial are now equatorial and vice versa.

The whole inversion process can be broken down into the conformations shown below. The green arrows show the direction in which the individual carbon atoms should move in order to get to the next conformation.



The energy profile (below) for this ring inversion shows that the half-chair conformation is the energy maximum on going from a chair to a twist-boat. The true boat conformation is the energy maximum on interchanging between two mirror-image twist-boat conformers, the second of which is converted to the other chair conformation through another half-chair.



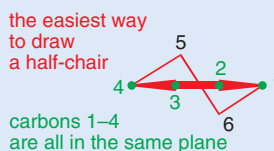
■ Make a model of cyclohexane and try the ring inversion for yourself.

Interactive conformations of cyclohexane

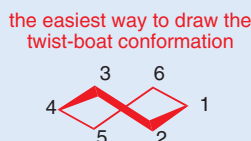
■ This would be a good point to remind you again of Chapter 12. This energy profile shows the conversion of one chair to another via two twist-boat *intermediates* (local energy minima). In between the energy minima are energy maxima, which are the *transition states* for the process. The progress of the ring-flipping ‘reaction’ is shown along an arbitrary ‘reaction coordinate’.

Drawing other cyclohexane conformations

In the **half-chair** conformation of cyclohexane, four adjacent carbon atoms are in one plane with the fifth above this plane and the sixth below it. You will meet this conformation again later—it represents the energy minimum for cyclohexene, for example.



There are also a number of ways of drawing a twist-boat conformer but the easiest is this:



It's clear from the diagram that the barrier to ring inversion of cyclohexane is 43 kJ mol^{-1} , a rate at 25°C of about $2 \times 10^5 \text{ s}^{-1}$. Ring inversion also interconverts the axial and equatorial protons, so these are also exchanging at a rate of $2 \times 10^5 \text{ s}^{-1}$ at 25°C —too fast for them to be detected individually by NMR, which is why they appear as an averaged signal.

Rates and spectroscopy

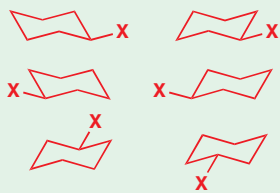
NMR spectrometers behave like cameras with a shutter speed of about $1/1000 \text{ s}$. Anything happening faster than that, and we get a blurred picture; things happening more slowly give a sharp picture. In fact, a more exact number for the 'shutter speed' of an NMR machine (not a real shutter speed—just figuratively speaking!) is given by the equation

$$k = \pi \Delta\nu / \sqrt{2} = 2.22 \times \Delta\nu$$

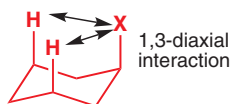
where k is the fastest exchange rate that still gives individual signals and $\Delta\nu$ is the separation of those signals in the NMR spectrum measured in hertz. For example, on a 400 MHz spectrometer, two signals separated by 0.5 ppm are 100 Hz apart, so any process exchanging with a rate slower than 222 s^{-1} will still allow the NMR machine to show two separate signals; if they exchange with a rate faster than 222 s^{-1} only an averaged signal will be seen.

The equation above holds for any spectroscopic method, provided we think in terms of differences between signals or peaks measured in hertz. So, for example, a difference between two IR absorptions of 100 cm^{-1} can be represented as a wavelength of 0.01 cm ($1 \times 10^{-4} \text{ m}$) or a frequency of $3 \times 10^{12} \text{ s}^{-1}$. IR can detect changes happening a lot faster than NMR can—its 'shutter speed' is of the order of one-trillionth of a second.

■ In a monosubstituted cyclohexane there is only one type of equatorial conformer and one type of axial conformer. Convince yourself that these drawings are exactly the same conformation just viewed from different vantage points:



➡ We discussed the relationship between energy differences and equilibrium constants in Chapter 12.



Substituted cyclohexanes

In a monosubstituted cyclohexane, there can exist two different chair conformers: one with the substituent axial, the other with it equatorial. The two chair conformers will be in rapid equilibrium (by the process we have just described) but they will not have the same energy. In almost all cases, *the conformer with the substituent axial is higher in energy*, which means there will be less of this form present at equilibrium.

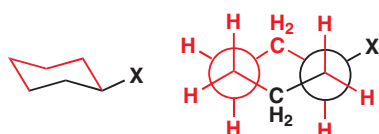


For example, in methylcyclohexane ($\text{X} = \text{CH}_3$), the conformer with the methyl group axial is 7.3 kJ mol^{-1} higher in energy than the conformer with the methyl group equatorial. This energy difference corresponds to a 20:1 ratio of equatorial:axial conformers at 25°C .

There are two reasons why the axial conformer is higher in energy than the equatorial conformer. The first is that the axial conformer is destabilized by the repulsion between the axial group X and the two axial hydrogen atoms on the same side of the ring. This interaction is known as the 1,3-diaxial interaction. As the group X gets larger, this interaction becomes more severe and there is less of the conformer with the group axial. The second reason is that

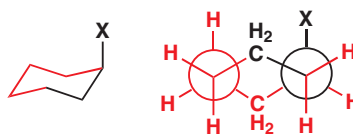
in the equatorial conformer the C–X bond is anti-periplanar to two C–C bonds, while, for the axial conformer, the C–X bond is synclinal (gauche) to two C–C bonds.

equatorially substituted cyclohexane:



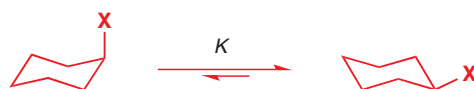
the black bonds are anti-periplanar
(only one pair shown for clarity)

axially substituted cyclohexane:



the black bonds are synclinal (gauche)
(only one pair shown for clarity)

The table shows the preference of a number of substituted cyclohexanes for the equatorially substituted conformer over the axially substituted conformer at 25 °C.

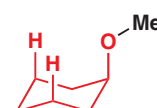
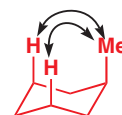


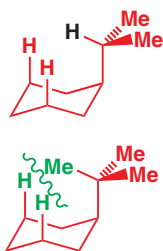
$$K = \frac{\text{concentration of equatorial conformer}}{\text{concentration of axial conformer}}$$

X	Equilibrium constant, K	Energy difference between axial and equatorial conformers, kJ mol^{-1}	Percentage with substituent equatorial
H	1	0	50
OMe	2.7	2.5	73
Me	19	7.3	95
Et	20	7.5	95
<i>i</i> -Pr	42	9.3	98
<i>t</i> -Bu	>3000	>20	>99.9
Ph	110	11.7	99

Note the following points (also referred to in Chapter 12).

- The three columns in the table are three different ways of expressing the same information. However, just looking at the percentages column, it is not immediately obvious to see how much more of the equatorial conformer there is—after all, the percentages of equatorial conformer for methyl, ethyl, isopropyl, *t*-butyl, and phenylcyclohexanes are all 95% or more. Looking at the equilibrium constants gives a much clearer picture.
- The amount of equatorial conformer present does increase in the order $\text{Me} < \text{Et} < i\text{-Pr} < t\text{-Bu}$, but perhaps not quite as expected. The ethyl group *must* be physically larger than a methyl group but there is hardly any difference in the equilibrium constants. The increase in the proportion of equatorial conformer on going from Et to *i*-Pr is only a factor of two, but for *t*-butylcyclohexane it is estimated that there is about 3000 times more of the equatorial conformer than the axial conformer.
- The same anomaly occurs with the methoxy group—there is a much greater proportion of the conformer with a methoxy group axial than with a methyl group axial. This is despite the fact that a methoxy group is physically larger than a methyl group.
- The equilibrium constant does not depend on the actual size of the substituent, but rather its interaction with the neighbouring axial hydrogens. In the axial conformer of methylcyclohexane there is a direct interaction between the methyl group and the axial hydrogen atoms.
- In the case of the methoxy group, the oxygen acts as link and removes the methyl group away from the ring, lessening the interaction.



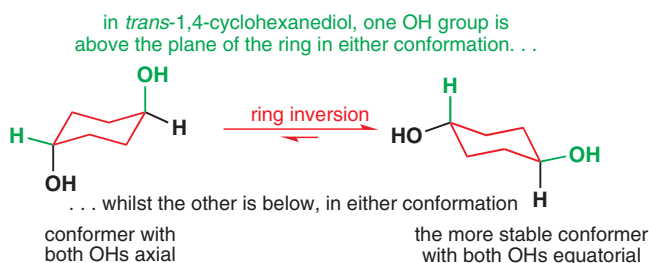
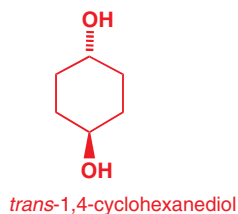
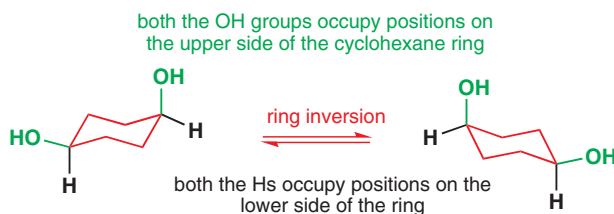
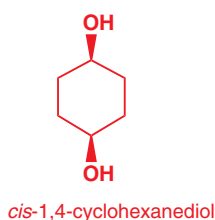


- The groups Me, Et, *i*-Pr, and *t*-Bu all need to point some atom towards the other axial hydrogens, and for Me, Et, and *i*-Pr this can be H.
- Only for *t*-Bu must a methyl group be pointing straight at the axial hydrogens, so *t*-Bu has a much larger preference for the equatorial position than the other alkyl groups. In fact, the interactions between an axial *t*-Bu group and the axial hydrogen atoms are so severe that the group virtually always stays in the equatorial position. As we shall see later, this can be very useful.

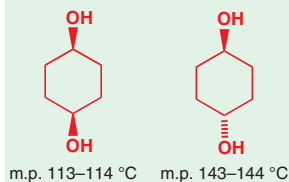
What happens with more than one substituent on the ring?

When there are two or more substituents on the ring, stereoisomerism is possible. For example, there are two isomers of 1,4-cyclohexanediol—in one (the *cis* isomer) both the substituents are either above or below the cyclohexane ring; in the other (the *trans* isomer) one hydroxyl group is above the ring whilst the second is below. For a *cis* 1,4-disubstituted cyclohexane with both the substituents the same, ring inversion leads to a second identical conformation, while for the *trans* configuration there is one conformation with both groups axial and one with both groups equatorial.

■ Ring inversion interconverts all of the axial and equatorial substituents, but it does not change which face of the ring a substituent is on. If an equatorial substituent starts off above the ring (that is, 'up' relative to its partner on the same C atom) it will end up above the ring, but now axial. Axial and equatorial are *conformational* terms; which side of the ring a substituent is on depends on the compound's *configuration*.



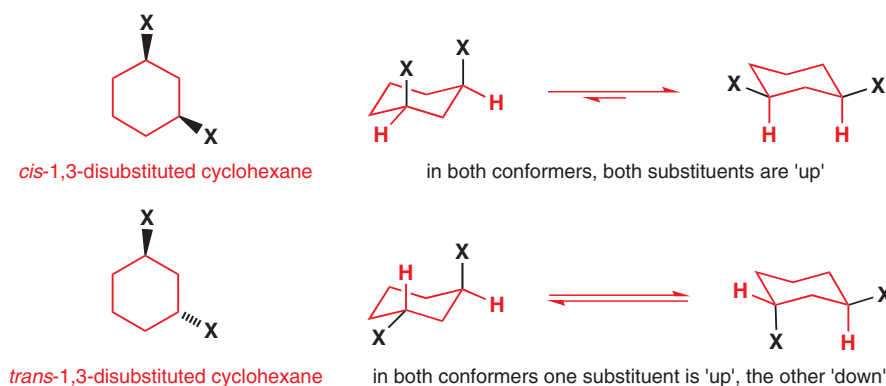
■ The *cis* and *trans* compounds are different diastereoisomers. Consequently, they have different chemical and physical properties and cannot interconvert simply by rotating bonds.



This contrasts with the two conformers of *trans*-1,4-hydroxycyclohexane (diaxial or diequatorial), which rapidly interconvert at room temperature without breaking any bonds.

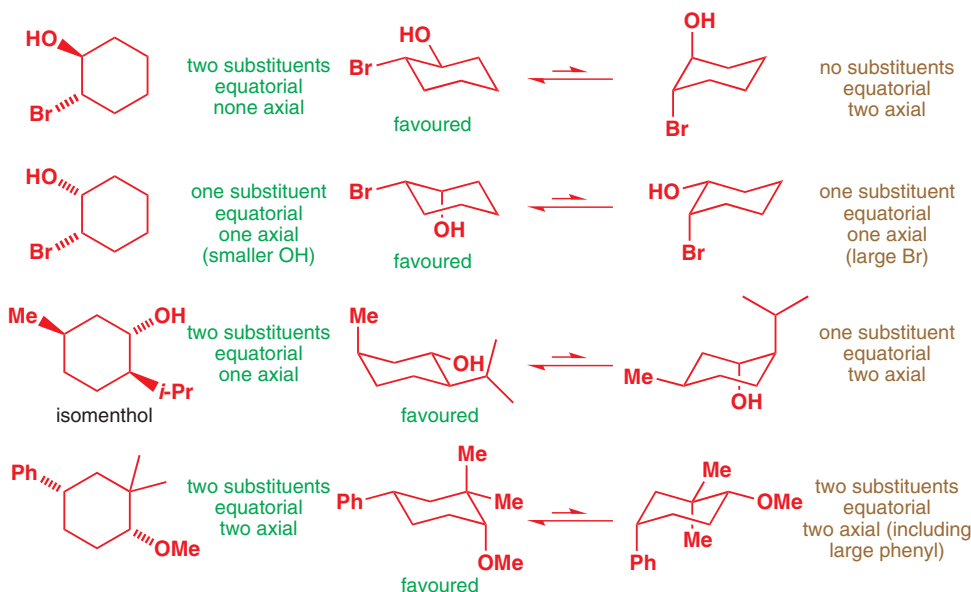
The chair-structure diagrams contain much more information than the simple 'hexagon' diagrams that we have used up to now. The former show both configuration and conformation—they show which stereoisomer (*cis* or *trans*) we are talking about and also (for the *trans* compound) the conformation adopted (diaxial or the more stable diequatorial). In contrast, the simpler hexagon diagrams carry no information about the conformation—only information about which isomer we are dealing with. This can be useful because it enables us to talk about one configuration of a compound without specifying the conformation. When you are solving a problem requiring conformational diagrams to predict the configuration of a product, always start and finish with a configurational (hexagon) drawing.

The chair conformer of *cis*-1,4-disubstituted cyclohexane has one substituent equatorial, the other axial. This will not necessarily be the case for other substitution patterns, for example the chair conformer of a *cis*-1,3-disubstituted cyclohexane has either both substituents axial or both equatorial. Remember, the '*cis*' and '*trans*' prefixes merely indicate that both groups are on the same 'side' of the cyclohexane ring. Whether the substituents are both axial/equatorial or one axial and the other equatorial depends on the substitution pattern. Each time you meet a molecule, draw the conformation or make a model to find out which bonds are axial and which are equatorial.



■ It is not always easy to decide if an equatorial substituent is 'up' or 'down'. The key is to compare it with its axial partner on the same C atom—axial substituents very clearly point 'up' or 'down'. If the axial partner is 'up', the equatorial substituent must be 'down' and vice versa.

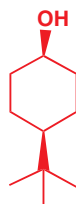
What if the two substituents on the ring are different? For the *cis*-1,3-disubstituted example above, there is no problem because the favoured conformation will still be the one that places these two different substituents equatorial. But when one substituent is axial and the other equatorial (as they happen to be in the *trans* diastereoisomer above) the preferred conformation will depend on what those substituents are. In general, the favoured conformation will place the maximum number of substituents equatorial. If both conformations have the same number of equatorial substituents, the one with the larger substituent equatorial will win out, and the smaller group will be forced to be axial. Various possibilities are included in the examples below.



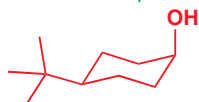
This is only a guideline, and in many cases it is not easy to be sure. Instead of concerning ourselves with these uncertainties, we shall move on to some differentially substituted cyclohexanes for which it is absolutely certain which conformer is preferred.

Locking groups—*t*-butyl groups, decalins, and steroids

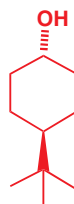
We have already seen how a *t*-butyl group always prefers an equatorial position in a ring. This makes it very easy to decide which conformation the two different compounds below will adopt.

cis-4-*t*-butylcyclohexanol

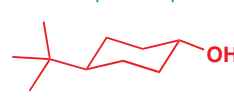
in the *cis* diastereoisomer, the hydroxyl group is forced into an axial position



in both compounds, the *t*-butyl group is equatorial

trans-4-*t*-butylcyclohexanol

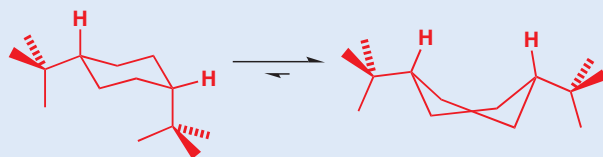
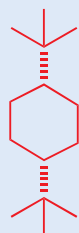
in the *trans* diastereoisomer, the hydroxyl group is forced into an equatorial position



in both compounds, the *t*-butyl group is equatorial

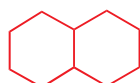
Cis-1,4-di-*t*-butylcyclohexane

An axial *t*-butyl group really is very uncomfortable. In *cis*-1,4-di-*t*-butylcyclohexane, one *t*-butyl group would be forced axial if the compound existed in a chair conformation. To avoid this, the compound prefers to pucker into a twist-boat so that the two large groups can both be in equatorial positions (or 'pseudoequatorial', since this is not a chair).

cis-1,4-di-*t*-butylcyclohexane

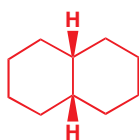
the twist-boat conformer (with both *t*-butyl groups in pseudoequatorial positions) is lower in energy than the chair conformer.

Decalins

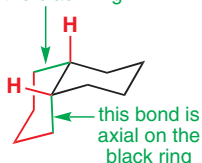


decalin

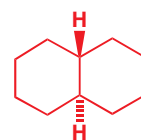
It is also possible to lock the conformation of a cyclohexane ring by joining another ring to it. Decalin is two cyclohexane rings fused at a common C–C bond. Two diastereoisomers are possible, depending on whether the hydrogen atoms at the ring junction are *cis* or *trans*. For *cis*-decalin, the second ring has to join the first so that it is axial at one point of attachment and equatorial at the other; for *trans*-decalin, the second ring can be joined to the first in the equatorial position at both attachment points.

*cis*-decalin

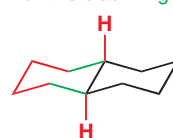
this bond is equatorial on the black ring



this bond is axial on the black ring

*trans*-decalin

both green bonds are equatorial on the black ring

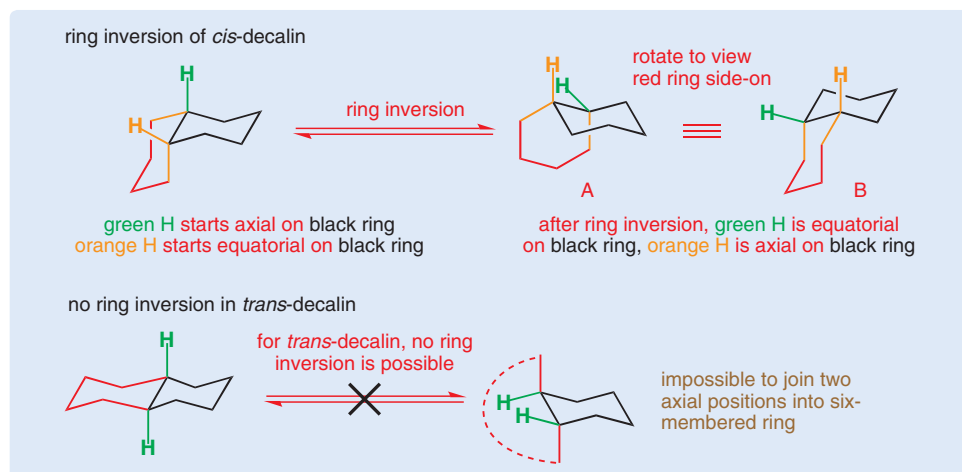


Interactive conformations of decalins

When a cyclohexane ring inverts, the substituents that were equatorial become axial and vice versa. This is fine for *cis*-decalin, which has an axial–equatorial junction, but it means that ring inversion is not possible for *trans*-decalin. For *trans*-decalin to invert, the junction would have to become axial–axial, and it's not possible to link the axial positions to form a six-membered ring. *Cis*-decalin, on the other hand, ring inverts easily.

Flipping *cis*-decalin: not so difficult once you know how

If you find it hard to visualize the ring inversion of *cis*-decalin, you are not alone! The best way to think about it is to ignore the second ring until the very end: just concentrate on what happens to one ring (black in this diagram), the hydrogens at the ring junction, and the (orange) bonds next to these hydrogens that form the 'stumps' of the second ring. Flip the black ring, and the 'stumps', and the hydrogens swap from axial to equatorial and vice versa. Draw the result, but don't fill in the second ring yet or it will usually just come out looking like a flat hexagon (as in diagram A). Instead, rotate the complete (black) ring 60° about a vertical axis so that both of the orange 'stumps' can form part of a chair, which can now be filled in (diagram B). To make a chair (and not a hexagon) they must be pointing in a convergent direction, as the orange bonds are in B but not A.



Interactive ring inversion of *cis*-decalin

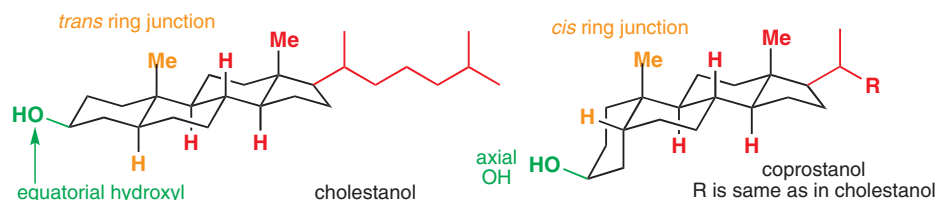
Steroids

Steroids are an important class of compounds occurring in all animals and plants, which have many important functions from regulating growth (anabolic steroids) and sex drive (all sex hormones are steroids) to acting as a self-defence mechanism in plants, frogs, and even sea cucumbers. A steroid is defined by its structure: all steroids contain a basic carbon framework consisting of four fused rings—three cyclohexane rings and one cyclopentane ring—labelled and joined together as shown in the margin.

Just as in the decalin system, each ring junction could be *cis* or *trans*, but it turns out that all steroids have all *trans*-junctions except where rings A and B join, which is sometimes *cis*. Examples are cholesterol (all *trans*) and coprostanol (A and B fused *cis*).



the steroid skeleton



Because steroids (even those with a *cis* A–B ring junction) are essentially substituted *trans*-decalins, they can't ring-flip. This means, for example, that the hydroxyl group in cholesterol is held equatorial on ring A while the hydroxyl group in coprostanol is held axial on ring A. The steroid skeleton really is remarkably stable—samples of sediment 1.5×10^9 years old have been found to contain steroids still with the same ring-junction stereochemistry.

Axially and equatorially substituted rings react differently

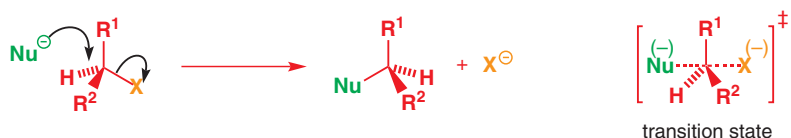
We shall be using ring structures throughout the rest of the book, and you will learn how their conformation affects their chemistry extensively. In many reactions of six-membered rings, the outcome may depend on whether a functional group is axial or equatorial. We shall conclude this chapter with two examples in which a functional group will be held in its axial or equatorial position by 'locking' the ring using a *t*-butyl group or a fused ring system such as *trans*-decalin.

In the last chapter we looked at two mechanisms for nucleophilic substitution: S_N1 and S_N2 . We saw that the S_N2 reaction involved an inversion at the carbon centre. Recall that the incoming nucleophile had to attack the σ^* orbital of the C–X bond. This meant that it had to approach the leaving group directly from behind, leading to inversion of configuration.

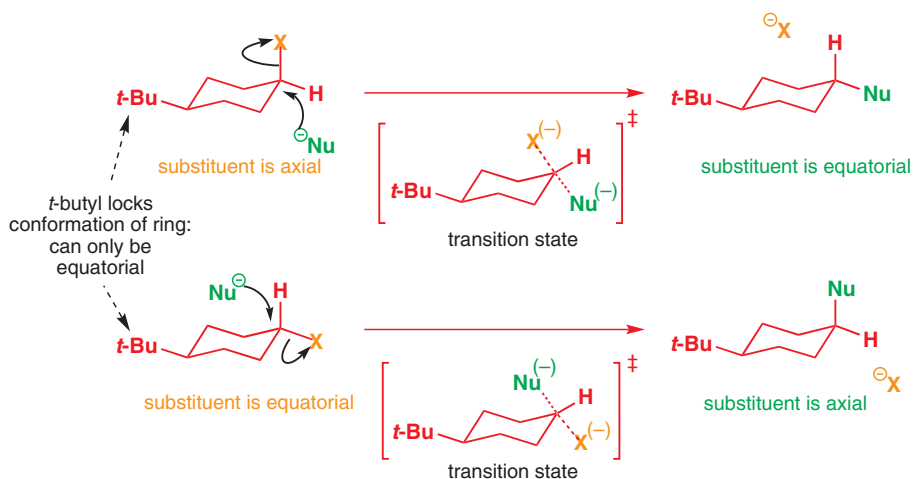
It was a desire to explain the reactions of steroids that led Sir Derek Barton (1918–98) to discover, in the 1940s and 1950s, the principles of conformational analysis described in this chapter. It was for this work that he shared the Nobel Prize for Chemistry in 1969.

■ These locking groups work because the *t*-butyl group can never be axial (p. 375) and the *trans*-decalin can never ring-flip (p. 378).

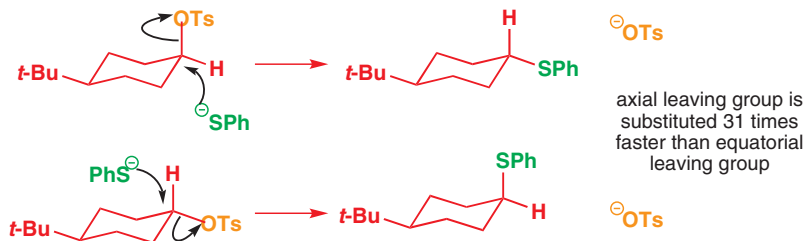
inversion during nucleophilic substitution at saturated carbon



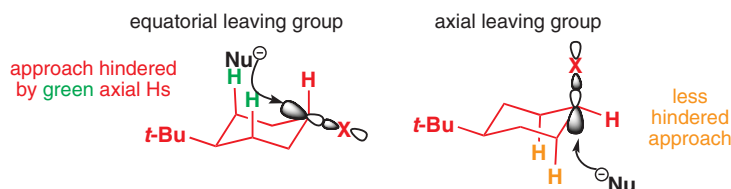
What do you think would happen if a cyclohexane derivative underwent an S_N2 reaction? If the conformation of the molecule is fixed by a locking group, the inversion mechanism of the S_N2 reaction means that, if the leaving group is axial, then the incoming nucleophile will end up equatorial and vice versa.



Substitution reactions are not very common for substituted cyclohexanes. The electrophilic carbon in a cyclohexane ring is a secondary centre—in the last chapter we saw that secondary centres do not react well via either S_N1 or S_N2 mechanisms (p. 347). To encourage an S_N2 mechanism, we need a good attacking nucleophile and a good leaving group. One such example is shown—the substitution of tosylate by PhS^- .

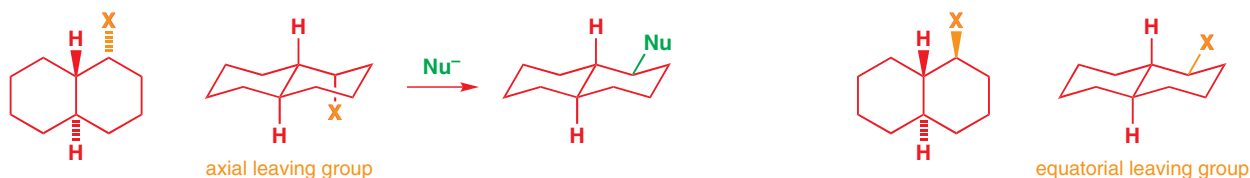


The substitution of an axial substituent proceeds faster than the substitution of an equatorial substituent. There are several contributing factors making up this rate difference, but the most important is the direction of approach of the nucleophile. The nucleophile must attack the σ^* of the leaving group, that is, directly behind the C–X bond. In the case of an equatorially substituted compound, this line of attack is hindered by the (green) axial hydrogens—it passes directly through the region of space they occupy. For an axial leaving group, the direction of attack is parallel with the (orange) axial hydrogens anti-periplanar to the leaving group, and approach is much less hindered.



We must assume that this holds even for simple unsubstituted cyclohexanes, and that substitution reactions of cyclohexyl bromide, for example, occur mainly on the minor, axial conformer. This slows down the reaction because before it can react, the prevalent equatorial conformer must first flip to the axial.

If this flip to an axial leaving group is not possible, it may be that the reaction just won't happen at all. This is exactly what happens in a *trans*-decalin. There are two diastereoisomers of this simple substituted *trans*-decalin: one with an equatorial and one with an axial leaving group (X could be Br, OTs, etc.).



Attack by a nucleophile on the compound with the axial leaving group is straightforward. The nucleophile can approach along the axis of the C–X bond and normal S_N2 reaction occurs with inversion—the product is the equatorial compound. The equatorial leaving group, on the other hand, would require the nucleophile to approach through the middle of the molecule and that cannot be achieved. A totally different reaction occurs—a rearrangement that you will meet in Chapter 36.

To conclude...

You may wonder why we have spent most of this chapter looking at six-membered rings, ignoring other ring sizes almost totally. Apart from the fact that six is the most widespread ring size in organic chemistry, the reactions of six-membered rings are also the easiest to explain and to understand. The conformational principles we have outlined for six-membered rings (relief of ring strain, staggered favoured over eclipsed, equatorial favoured over axial, direction of attack) hold, in modified form, for other ring sizes as well. These other rings are less well behaved than six-membered rings because they lack the well-defined strain-free conformations that cyclohexane is blessed with. We shall now leave stereochemistry in rings for some time, but we come back to these more difficult rings—and how to tame them—in two chapters on stereochemistry in cyclic compounds, Chapters 31 and 32.

Further reading

A major resource for conformation is E. L. Eliel and S. H. Wilen, *Stereochemistry of Organic Compounds*. Wiley, New York, 1994.

For a more detailed analysis of reasons for the conformational preferences of alkanes, see V. Pophristic and L. Goodman, *J. Phys. Chem. A*, 2002, 106, 1642–1646.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>